

Spoken Tutorial – Understanding AI, Machine Learning, and Deep Learning

Title: Understanding AI, Machine Learning, and Deep Learning

Module	Introduction to AI
Episode	1
Learning Objective	At the end of this tutorial learner will be able to 1. Understand the core definition and goal of Artificial Intelligence. 2. Explain the concept of Machine Learning and differentiate it from traditional programming. 3. Identify real-world applications of Machine Learning. 4. Define Deep Learning and describe its relationship to Machine Learning. 5. Recognize practical examples of Deep Learning in action. 6. Compare and contrast Artificial Intelligence, Machine Learning, and Deep Learning.
Approx. Duration	12 minutes

Outline	• At its core, Artificial Intelligence is a broad field of computer science focused on a simple goal: building smart machines capable of performing tasks that typically require human intelligence. • This could be anything from understanding language and recognizing images to solving complex problems. • The idea isn't new; it has been a dream of scientists since the 1950s, starting with pioneers who asked, "Can machines think?" AI is the entire universe of making computers intelligent. • What is Machine Learning (ML)? • Machine Learning is a type of AI. Instead of giving a computer a detailed list of rules to follow (traditional programming), we give it a lot of data and let it learn the rules for itself. It's all about recognizing patterns from examples. • Let's use an analogy: • Traditional Programming (Rules First): Imagine you want to program a computer to identify a cat. You'd have to write specific rules like: "IF it has pointy ears, AND it has whiskers, AND it has fur, AND it has four legs, THEN it is a cat." This is fragile what if the cat's ears are folded? • Machine Learning (Data First): Instead, you show the computer thousands of pictures labeled "cat." The ML model analyzes all these images and learns the common patterns and features that define a cat on its own. It builds its own internal "rules" that are much more flexible and accurate. • What is Deep Learning? • Deep Learning is a super-powered, more advanced version of Machine Learning. It uses a structure called a neural network, which is inspired by the human brain. • These networks have many layers of "neurons" stacked on top of each other and because there are many layers, we call it "deep." • This deep structure allows it to learn much more complex patterns from data than traditional ML. • Deep learning is the magic behind self-driving cars recognizing pedestrians, voice assistants understanding your commands, and the generative AI we'll discuss next.
Meta Tags	AI, Machine Learning, Deep Learning, Artificial Intelligence, ML, DL, Neural Networks, Spoken Tutorial, Introduction to AI, Tech Concepts
Pre-requisite Tutorial	Basic computer literacy and an interest in technology.

Script

Visual Cue	Narration
	Hello and welcome to this Spoken Tutorial. Here, we will learn about Artificial Intelligence, Machine Learning, and Deep Learning. We will understand these important topics together.

	After this tutorial, you will know what Artificial Intelligence is. You will also understand Machine Learning and how it is different from old ways of programming. We will also learn about Deep Learning and its uses.
	To follow this tutorial, you need a computer with internet. You also need a web browser. You do not need to install any special software for this lesson.
	To do this tutorial, you should know how to use a computer. You should also be interested in new technology. You do not need to know programming or tough maths.
A futuristic robot thinking, with gears and light bulbs representing intelligence, set against a backdrop of computer science elements.	Artificial Intelligence, or AI, is a big part of computer science. Its main aim is to make smart machines. These machines can do tasks that usually need human thinking. Have you ever thought, 'Can machines think like us'?
A timeline showing the evolution of AI, from 1950s concepts to modern applications, with icons representing different AI tasks like language processing and image recognition.	AI can help machines understand language or see images. It can also solve difficult problems. This idea is not new; scientists have dreamt of it since the 1950s. They asked, 'Can machines think?' AI is the whole world of making computers smart.
A computer learning from a large dataset, with data flowing into a processing unit, and patterns emerging as a result of the learning process.	Now, what is Machine Learning? We call it ML. ML is one part of AI. Instead of telling the computer every rule, we give it a lot of data. The computer then learns the rules on its own. It finds patterns from many examples.
A cartoon cat with distinct features, next to a flowchart showing rigid 'IF-THEN' rules for identification, highlighting the limitations of rule-based systems.	Let us understand this with an example. Imagine you want to teach a computer to find a cat. In old programming, you write rules. For example: 'If it has pointy ears and whiskers, then it is a cat.' But what if the cat's ears are folded? This method is not strong.
A collage of diverse cat images, with an arrow pointing to a machine learning model that is processing them and learning to identify a cat flexibly.	Machine Learning works in another way. You show the computer many pictures of cats. The computer looks at all these pictures. It learns what makes a cat a cat. It makes its own rules that are much better and more correct.

A stylized representation of a human brain next to a simplified neural network diagram, illustrating the conceptual inspiration for neural networks.	Now, let us talk about Deep Learning. Deep Learning is a very strong and advanced type of Machine Learning. It uses something called a neural network. This network works a little like the human brain.
A multi-layered neural network diagram, emphasizing the many stacked layers of interconnected 'neurons' to represent depth.	These networks have many layers of 'neurons'. They are like many layers on top of each other. Because of these many layers, we call it 'deep'. This deep structure helps it learn very complex patterns from data. It learns better than old ML methods.
A composite image showing a self-driving car detecting a pedestrian, a person interacting with a voice assistant, and a visual representation of generative AI creating art.	Deep Learning helps self-driving cars see people on the road. It helps voice assistants understand what you say. It also helps with generative AI. We will talk about this later. See how Deep Learning makes our lives easier.
A Venn diagram showing AI as the largest encompassing circle, ML as a smaller circle inside AI, and DL as an even smaller circle nested inside ML.	Let us quickly look again. AI is the biggest field that wants to make machines smart like humans. Machine Learning is a part of AI where computers learn from data. Deep Learning is a special part of Machine Learning. It uses many-layered neural networks for very difficult tasks.
	In this tutorial, you learned the main differences. These are between Artificial Intelligence, Machine Learning, and Deep Learning. We saw that AI wants to make smart machines. ML helps machines learn from data. And DL uses deep neural networks to find complex patterns.
	Here is an assignment for you. Think of one real-world use for AI, ML, and DL each. Choose uses that we did not talk about in this tutorial. Write them down. Think about how each technology helps in that use.
	This Spoken Tutorial is brought to you by EduPyramids Educational Services Private Limited. It is also supported by SINE, IIT Bombay. Thank you for joining us!